

THE CODE OF LIFE

The science of DNA is a long story, featuring thousands of plants, animals, people, and microbes. Here are some of the key stepping-stones in the quest to read, to understand, and even to change the molecule that codes all life on Earth.

Double helix

Rosalind's image helps James Watson and Francis Crick to figure out the structure of DNA—a double helix. This is a big clue to how it works.



Marshall Nirenberg, Har Khorana, and Severo Ochoa decipher the code inside DNA—it instructs cells to build proteins.

Scientists develop ways to read the order of chemicals called nucleotides inside DNA—this is called sequencing.

1953

1966

1975/77

A disease called sickle cell anemia is shown to be caused by an abnormal gene that can be passed from parent to child.

1978

Gene transfer

Scientists figure out how to take a gene from one living thing and add it to the DNA of a mouse or a fruit fly. A gene is a stretch of DNA that tells a cell how to make a certain protein.

Insulin is manufactured by adding the gene that makes it to bacteria, turning the bacteria into tiny insulin factories! Insulin can treat a disease called diabetes.

1981

1982

PCR

A method for making millions of copies of a stretch of DNA is developed, called a polymerase chain reaction (PCR). It makes it much easier to sequence DNA and to figure out what genes do.

1983



2020

Rapid genome sequencing helps scientists understand the coronavirus that causes COVID-19. CRISPR gene-editing technology is used to help develop vaccines.

2020



CRISPR therapies

The gene editing therapy used for sickle cell disease—clustered regularly interspaced short palindromic repeats (CRISPR)—is used inside other patients' bodies. It treats illnesses that affect the eyes and liver.